



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
15-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

Time : 3 Hours

PAPER-1

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Spherical sound waves are emitted uniformly in all directions from a point source. The variation in sound level SL as a function of distance $r(> 0)$ from the source can be written as:- (where a and b are positive constant)

(A) $SL = -b \log r^a$

(C) $SL = a - b(\log r)^2$

(B) $SL = a - b \log r$

(D) $SL = a - \frac{b}{r^2}$
2. Steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent 0.02 kg at 15°C till the temperature of the calorimeter and its contents rises to 80°C . What is the mass of steam condensed? Latent heat of vaporization of water is 536 cal/gm.

(A) 0.23 kg

(B) 0.13 kg

(C) 0.05 kg

(D) 0.42 kg
3. A metal plate when exposed to light of wavelength 310 nm, photoelectrons are ejected. When a retarding electric field of intensity $E = 100 \text{ N/C}$ is applied, none of the photoelectrons can move away from the plate farther than 1 cm. Which of the following is a correct value for the threshold wavelength?

(A) 413.33 nm

(B) 620 nm

(C) 496 nm

(D) 826.66 nm

Space for Rough Work

4. A stream of photons is made to pass through a gaseous sample consisting of identically excited hydrogen like ions. If every photon has energy $144E_0/225$, ten distinct lines are observed in the emission spectrum of the sample. Some of these lines correspond to energy lower than, some higher than and some equal to the energy of the incident photons. Here E_0 is the ionization energy of a hydrogen atom in ground state. Identify the element.
- (A) Flourine (B) Beryllium (C) Boron (D) Lithium
5. One slit of a double slit experiment is covered by a thin glass plate of refractive index 1.4 and the other by a thin glass plate of refractive index 1.7. The point on the screen, where central bright fringe was formed before the introduction of the glass sheets, is now occupied by the 5th bright fringe. Assuming that both the glass plates have same thickness and wavelength of light used is $4800 \text{ \AA}$, find their thickness.
- (A) $6 \mu\text{m}$ (B) $8 \mu\text{m}$ (C) $4 \mu\text{m}$ (D) $2 \mu\text{m}$
6. Find the initial momentum of electron if the momentum of electron is changed by p_m and the De Broglie wavelength associated with it changes by 0.50 %
- (A) $50 p_m$ (B) $100 p_m$ (C) $200 p_m$ (D) $250 p_m$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

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- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

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7. An ac ammeter, an ac voltmeter and a resistor are connected in series and the combination is connected across a sinusoidal voltage source. The ac ammeter and the ac voltmeter are purely resistive in nature. The readings are A and V respectively. If a capacitor is now connected across the voltmeter

(A) A will increase

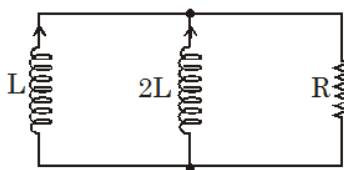
(B) A will decrease

(C) V will increase

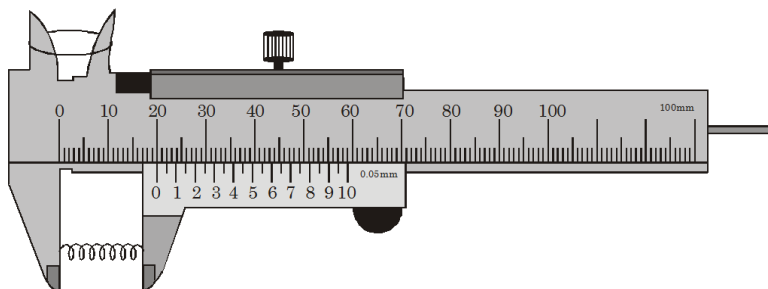
(D) V will decrease

Space for Rough Work

8. A parallel circuit consists of two ideal inductances of 1 H and 2 H and a resistance of 100Ω . At an instant the same current of 3 A is flowing in both the inductances in the same direction as shown. Choose the correct statement(s)



- (A) Total charge of 0.04 C will pass through the resistor
 (B) In the steady state, the current in the resistor will be zero
 (C) In the steady state, the current through each of the inductor will be 1 A and will be in the anticlockwise direction
 (D) Initial potential difference across each of the inductor is 600 V .
9. A rubber band of circumference 3.9 cm and force constant 100 N/m has negligible thickness. It is tied to the upper jaws of vernier calipers as shown. The lower jaws are separated by a spring of force constant 200 N/m . The reading of the vernier calipers is 20 mm at the main scale and zero of vernier scale coincides with that scale. There is no zero error.



- (A) The natural length of the spring is 2.1 cm
 (B) The natural length of the spring is 2.05 cm
 (C) If the rubber band was looped in two loops and then tied to the upper jaws, the reading of the vernier calipers would be 11 msd and 0 of vernier scale coinciding
 (D) If the rubber band was looped in two loops and then tied to the upper jaws, the reading of the vernier calipers would be 10 msd and 8th vernier division coinciding

Space for Rough Work

10. Two particles of a medium in which a wave is propagating are located at $x_1 = 0$ and $x_2 = 1$ cm. The displacements of these particles is given by equation $y_1 = 2\sin 3\pi t$ cm, $y_2 = 2\sin(3\pi t - \frac{\pi}{8})$ cm. t is in seconds. The wave is known to travel in positive x direction.
- (A) The frequency of the wave is 1.5 Hz
 (B) The speed of the wave can be 16 cm/sec
 (C) The wavelength of the wave can be 16 cm
 (D) The value of displacement of particle at $x = 4$ cm at $t = 1$ sec is 2 cm
11. A solar cooker is a spherical shell of radius 30 cm blackened at its outer surface and kept in bright sunlight. Assume that it is completely filled with water. The shell is thin and can be assumed to be perfect conductor. The solar constant is 1536 W/m^2 . The surrounding is at 27°C . Assume that the heat exchange with the surroundings is only due to radiation. Take Stefan's constant as $6 \times 10^{-8} \text{ W/m}^2\text{K}^4$.
- (A) The steady state temperature of water is independent of the emissivity of the shell
 (B) The steady state temperature of the blackened shell is approximately 346 K
 (C) The steady state temperature of the blackened shell is approximately 282 K
 (D) If we have a larger sphere, steady state temperature would be higher.
12. In a standard YDSE setup, the fringe width is 6 mm. The intensity at the central maxima is 2 W/m^2 . At what distance from the central maxima will the intensity be 1 W/m^2 ?
- (A) 1.5 mm (B) 3 mm (C) 4.5 mm (D) 7.5 mm

Space for Rough Work

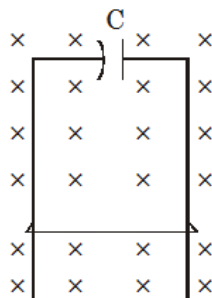
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
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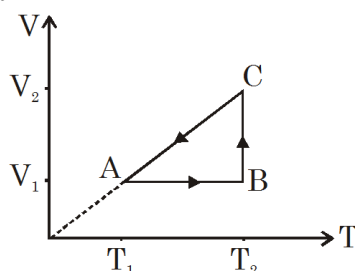
1. A wire of mass 20 gm and length 3 m can slide freely on a pair of smooth, vertical rails (figure). A magnetic field $B = 1$ T exists in the region in the direction perpendicular to the plane of the rails. The rails are connected at the top end by a capacitor of capacitance $2000\mu\text{F}$. The acceleration of the wire neglecting any electric resistance is _____ m/s^2 .



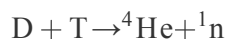
2. A mass of 1.00 kg is balanced on a massless meter stick hinged at the center by another unknown mass on the other side of the hinge. The least count of the meter scale is 1 mm. When 1 kg is at 14.0 cm, the other mass should be hanged at 59.0 cm to keep the meter scale horizontal. What is the percentage error in the measurement of the mass if known mass is known precisely?

Space for Rough Work

3. A steel rod rigidly clamped at one end and free at the other has longitudinal stationary waves set up in it and it vibrates in a mode in which there are eight displacement nodes other than the rigid end. The young's modulus for steel is 2.0×10^{11} Pa and its density is $8 \times 10^3 \text{ kg m}^{-3}$. The frequency of the source that sets the longitudinal waves in the rod is 10kHz. The length of the rod is _____ m
4. A screw gauge is made up of a material of thermal expansion coefficient $2 \times 10^{-5}/^\circ\text{C}$. It is designed to measure correctly at 0°C . It has 1mm as main scale division and 200 divisions on the circular scale. We want to measure the thickness of a bar at 100°C . The bar has a coefficient of thermal expansion $10^{-4}/^\circ\text{C}$. It measured 1 msd and 34 circular scale divisions at 0°C . What will be the new reading on circular scale?
5. Figure shows a process ABCA performed on one mole of an ideal monoatomic gas. Find the net heat given (in J) to the system during the cycle if $T_2 = 2 T_1$, $T_1 = 300 \text{ K}$, $R = 25/3 \text{ J/mol K}$, $\ln 2 = 0.7$.



6. The nuclei of deuterium D (^2H) and tritium T (^3H) can fuse according to the following reaction to produce a neutron and an alpha particle.



Calculate energy (in MeV) carried away by the neutron in the above nuclear reaction.

For an approximate estimation, assume the binding energies per nucleon of deuterium, tritium and helium to be 1.00 MeV, 2.80 MeV and 7.00 MeV respectively and neglect kinetic energy of the nuclei before the fusion.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 18)

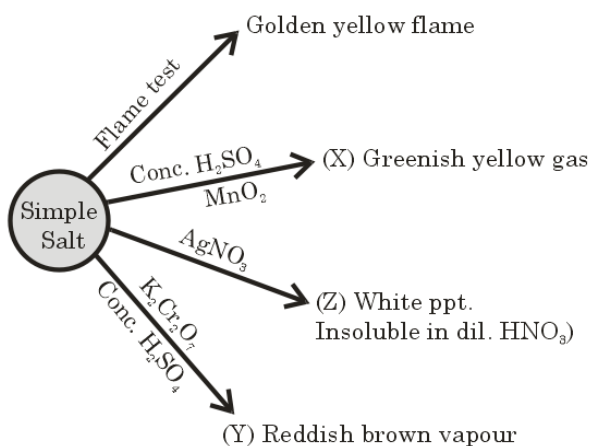
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1. Which of the statement is incorrect for the given paragraph

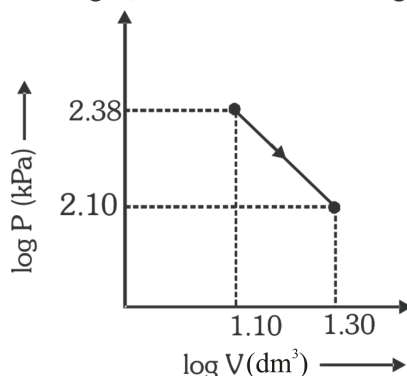


- (A) (X) can be formed by oxidation of HCl with O_2 in the presence of $CuCl_2$ as catalyst
- (B) When (Y) passed in NaOH solution yellow solution is formed
- (C) (Z) does not give chromyl chloride test
- (D) (X) when reacted with excess of NH_3 then one of the product is explosive in nature

Space for Rough Work

2. Which of the following statement is incorrect regarding the extraction of copper from low grade ore and scraps
- (A) Extracted by hydrometallurgy
- (B) Acids or bacteria are used in leaching process
- (C) H_2 can be used to recover copper from Cu^{2+} solution
- (D) Zn scrap is advisable to use in place of Fe scrap to recover copper from Cu^{2+} solution due to economical factor

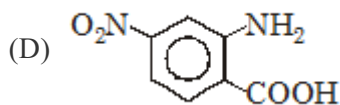
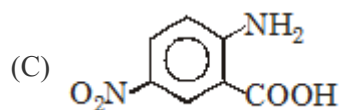
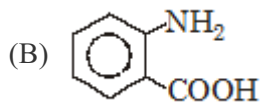
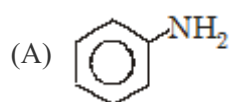
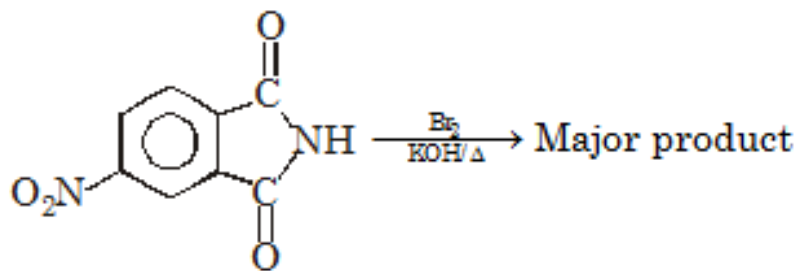
3. $\log P$ vs $\log V$ curve is plotted for an ideal gas, which is true for the gas ?



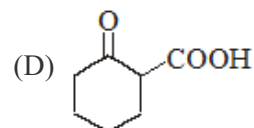
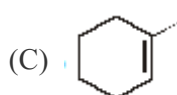
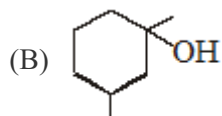
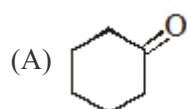
- (A) Monoatomic & undergoing adiabatic change
- (B) Monoatomic & undergoing an isothermal change
- (C) Diatomic & undergoing an adiabatic change
- (D) Triatomic & undergoing an isothermal change.
4. The conductance of 0.0015M aqueous solution of a weak monobasic acid was determined by using a conductivity cell consisting of platinised Pt electrodes. The distance between the electrodes is 120 cm with an area of cross section of 1cm^2 . The conductance was found to be $5 \times 10^{-7}\text{S}$. The pH of the solution is 4. The value of limiting molar conductivity (Λ_m°) of this weak monobasic acid in aq. solution is :-
- (A) $600\text{ S cm}^2\text{ mol}^{-1}$ (B) $400\text{ S cm}^2\text{ mol}^{-1}$
- (C) $300\text{ S cm}^2\text{ mol}^{-1}$ (D) $500\text{ S cm}^2\text{ mol}^{-1}$

Space for Rough Work

5. What will be the major product of the following reaction



6. Identify A
-
- Reaction scheme: Diethyl heptanedioate reacts with (1) EtONa , (2) $\text{H}_3\text{O}^+/\Delta$, and (3) $\text{CH}_3\text{MgI}/\text{H}^+/\Delta$ to form product A.



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

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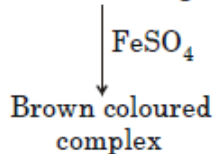
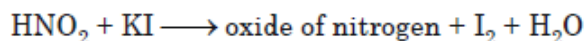
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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7. Consider the following reactions sequence.

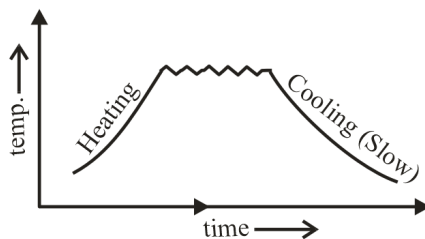


Choose the correct statement(s) about brown coloured complex

- (A) Magnetic moment of brown coloured complex is $\sqrt{15}$ B.M.
- (B) Brown coloured complex is coloured due to charge transfer spectra
- (C) Oxidation state of iron in brown coloured complex is +1
- (D) Hybridisation of brown coloured complex is sp^3d^2

Space for Rough Work

8. Consider the following graphical representation of heat treatment given to steel sample.



it cannot be the representation of :-

- (A) Quenching (B) Case hardening (C) Annealing (D) None of these
9. Select the correct statement(s)
- (A) If atom of tetrahedral void is displaced into octahedral void, then it is schottky defect.
- (B) Dislocation defect doesnot change density of crystal.
- (C) If radius of atom is $2\sqrt{2}\text{Å}$, then length of body diagonal in fcc unit cell formed by these atoms is $8\sqrt{3}\text{Å}$.
- (D) Hexagonal close packing has greater packing fraction than cubic close packing.

Space for Rough Work

10. For an ideal liquid - liquid solution of A($P_A^0 = 25$ torr) & B($P_B^0 = 50$ torr), select the correct option (s)

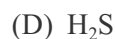
(A) $\frac{Y_A}{X_A}$ always less than $\frac{Y_B}{X_B}$

(B) $\frac{Y_A}{X_A}$ always more than $\frac{Y_B}{X_B}$

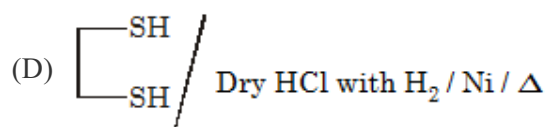
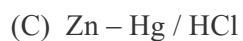
(C) $\frac{Y_A}{X_A} = \frac{Y_B}{X_B}$

(D) $\frac{Y_B}{X_A}$ may be greater, smaller or equal to $\frac{Y_A}{X_B}$

11. How many of the following can be used as R_1 $\text{CH}_3 - \text{COOH} \xrightarrow[\Delta]{R_1} \text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{O} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$



12. Which of the following reagents can be used to convert cyclohexanone into cyclohexane?



Space for Rough Work

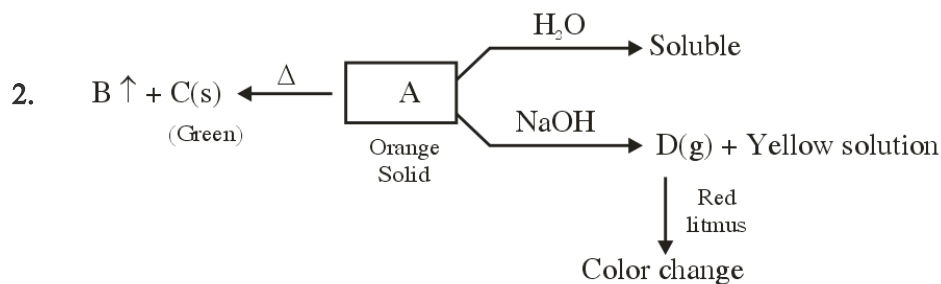
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1. Consider the following complex for given three statements :
- dichlorobis(ethylenediamine)chromium(III) chloride
- (i) Number of unpaired electrons in complex = (X)
- (ii) One mole of complex $\xrightarrow[\text{excess}]{\text{AgNO}_3}$ (Y) mole of AgCl.
- (iii) Total stereo isomeric forms of above complex = (Z) Find X + Y + Z



Number of correct statement(s) regarding A,B,C,D among the following

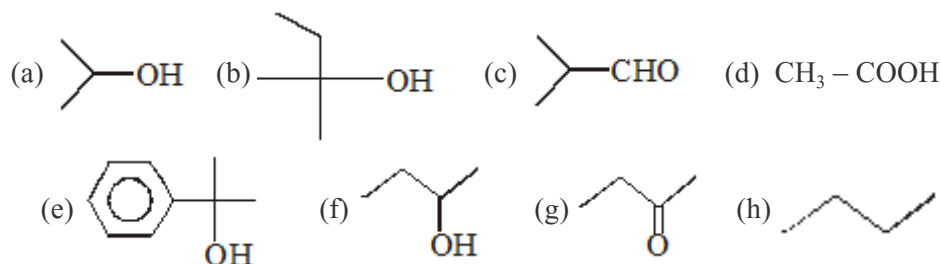
- (i) C is amphoteric
- (ii) Compound "A" contains X–O–X linkage
- (iii) Gas B is also released on heating nitrate of cation of A
- (iv) Anion of "A" has metal atom in d^3 hybridization.
- (v) D can symmetrically cleave Bridge Bond of Diborane.

Space for Rough Work

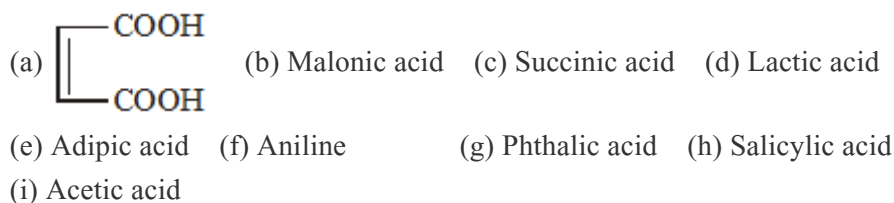
3. At 200°C, PCl_5 dissociates as $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$. It was found that the equilibrium vapours are 62 times as heavy as hydrogen. The percentage dissociation of PCl_5 at 200°C is.
[Atomic mass : P = 31, Cl = 35.5]

4. In order to cause coagulation of 100 ml arsenious sulphide sol, 111.7 ml of 2 m NaCl solution is used. What will be the flocculation value of NaCl ?

5. How many of the following can change the colour of acidic dichromate solution.



6. How many of the following will get decarboxylated on heating?



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PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Let $\vec{p}, \vec{q}, \vec{r}$ be three unit vectors such that $\vec{p} \times \vec{q} = \vec{r}$. If $\vec{a}$ is any vector such that $[\vec{a} \vec{q} \vec{r}] = 1$, $[\vec{a} \vec{r} \vec{p}] = 2$ and $[\vec{a} \vec{p} \vec{q}] = 3$, then $\vec{a} =$
 (A) $\vec{p} + 3\vec{q} + \vec{r}$ (B) $3\vec{p} + 2\vec{q} + \vec{r}$ (C) $\vec{p} - 2\vec{q} - 3\vec{r}$ (D) $\vec{p} + 2\vec{q} + 3\vec{r}$
2. Area bounded by $y = x^2 - 4$ & $y = 4 - x^2$ and the line $y = 4$ is -
 (A) $6\sqrt{6} - \frac{8}{3}$ (B) $\frac{64\sqrt{2}}{3} - \frac{64}{3}$ (C) $3\sqrt{3} + \frac{5}{4}$ (D) $6\sqrt{6} + \frac{5}{2}$
3. Let $f(x)$ be a twice differentiable function such that $f''(0) = 2$, then the value of $\lim_{x \rightarrow 0} \frac{f(5x) - 3f(3x) + 2f(2x)}{x^2}$ is
 (A) 0 (B) 3 (C) 6 (D) Does not exist

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4. If $\int \frac{\operatorname{cosec}^2 x}{(\operatorname{cosec} x + \cot x)^{9/2}} dx = (\operatorname{cosec} x - \cot x)^{7/2} \left(\frac{1}{\alpha} + \frac{(\operatorname{cosec} x - \cot x)^2}{11} \right) + C$

where C is constant of integration and $\alpha \in \mathbb{N}$, then α is -

- (A) 5 (B) 9 (C) 10 (D) 7

5. Let M be a 3×3 matrix satisfying $M^3 = O$. Then value of $\det \left(\frac{1}{2} M^2 + M + I \right)$ can not be

- (A) 0 (B) 1 (C) 2 (D) 3

6. $\int \frac{\sqrt{x}}{(1+x^2)^{7/4}} dx$ is equal to

(Where c is constant of integration)

(A) $\frac{2}{3} \frac{x^{3/2}}{(x^2+1)^{3/4}} + c$

(B) $\frac{1}{3} \frac{x^{3/2}}{(x^2+1)^{3/4}} + c$

(C) $\frac{1}{2} \frac{x^{1/2}}{(x^2+1)^{3/4}} + c$

(D) $\frac{x^{3/2}}{(x^2+1)^{3/4}} + c$

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SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. The curve(s) which satisfy the differential equation $x dy = y dx + \sin\left(\frac{1}{x}\right) dx$ is/are -

(A) $y = x \left(\cos\left(\frac{1}{x}\right) + 3 \right)$

(B) $y = 2x \left(2\cos^2 \frac{1}{2x} + \sin^2 \frac{1}{2x} \right)$

(C) $y = x^2 \left(\cos\left(\frac{1}{x}\right) + 3 \right)$

(D) $y = 2x \left(2\cos^2 \frac{1}{2x} - \sin^2 \frac{1}{2x} \right)$

8. Let f be a twice differentiable function on $[0,2]$ such that $f(0) = 0$, $f(1) = 2$, $f(2) = 4$, then-

(A) $f'(\alpha) = 2$ for some $\alpha \in (0,1)$

(B) $f''(x) = 0 \forall x \in (0,2)$

(C) $f'(\beta) = 2$ for some $\beta \in (1,2)$

(D) $f''(\gamma) = 0$ for some $\gamma \in (0,2)$

Space for Rough Work

9. Let $f(x) = 2 - |x - 1|$ & $g(x) = (x - 1)^2$, then -
- (A) area bounded by $f(x)$ & $g(x)$ is $\frac{7}{6}$
- (B) area bounded by $f(x)$ & $g(x)$ is $\frac{7}{3}$
- (C) area bounded by $f(x)$, $g(x)$ & x-axis is $\frac{5}{3}$
- (D) area bounded by $f(x)$, $g(x)$ & x-axis is $\frac{5}{6}$
10. Let $\vec{v}_1, \vec{v}_2, \vec{v}_3$ be three vectors such that $|\vec{v}_1|^2 = |\vec{v}_2|^2 = |\vec{v}_3|^2 = 3$ and $|\vec{v}_1 + \vec{v}_2 - \vec{v}_3|^2 + |\vec{v}_2 + \vec{v}_3 - \vec{v}_1|^2 + |\vec{v}_3 + \vec{v}_1 - \vec{v}_2|^2 = 36$, then
- (A) $\vec{v}_1 \cdot \vec{v}_2 + \vec{v}_2 \cdot \vec{v}_3 + \vec{v}_3 \cdot \vec{v}_1 = \frac{-9}{2}$
- (B) $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are coplanar vectors
- (C) $\vec{v}_1 \times \vec{v}_2 = \vec{v}_2 \times \vec{v}_3 = \vec{v}_3 \times \vec{v}_1 \neq \vec{0}$
- (D) $\vec{v}_1, \vec{v}_2, \vec{v}_3$ represent the sides of triangle.
11. If A and B are orthogonal matrices, then which of the following is/are also orthogonal matrix -
- (A) A^2
- (B) AB
- (C) A^2B^2
- (D) $A^T B$
12. If $P_n = \lim_{x \rightarrow 0^+} (\log(\tan x) + \log(\tan 2x) + \log(\tan 3x) + \dots + \log(\tan nx) - \log x^n)$ and $P_{256} - P_{255} = kP_m$, (where $k, m, n \in \mathbb{N}$), then-
- (A) $k = 8$
- (B) $k = 6$
- (C) $m = 2$
- (D) $m = 4$

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Given two sequences $a_n = \int_0^1 (1-x^2)^n dx$ & $b_n = \int_0^1 (1-x^3)^n dx$, ($n \in \mathbb{N}$), then the value of $\lim_{n \rightarrow \infty} (10\sqrt[n]{a_n} + 5\sqrt[n]{b_n})$ is equal to
2. Let λ_1 and λ_2 are order and degree of differential equation $\left(\frac{dy}{dx}\right)^{1/3} + \left(\frac{dy}{dx}\right)^{2/3} = x$, then $|(\lambda_1 - \lambda_2)|$ is
3. Point P(a, b, c) satisfies $a - b + c = 3$ and $a^2 + b^2 + c^2 = 4$. The point P traces a plane figure whose area is
4. The curve $y = e^{2x} - 18x + 15$ crosses the y-axis at P and has point of minima at Q. The area bounded by the curve and chord PQ is (use $\ln 3 = 1.1$)
5. Let B be a square matrix such that $|B| = 1$ and $2A + B = B^T - A^T$, then $\frac{1}{2}|A + B|$ is -
6. The value of $\left| \text{adj}\left(\frac{A}{3}\right)^{-1} \right|$, if $|A|_{3 \times 3} = 9$ is

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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